

Whisper Speech-to-Text Evaluation Report

Overview

This evaluation assesses OpenAI's Whisper speech-to-text (STT) API to determine its suitability for large-scale meeting transcription. The client processes **more than 1,000 hours of video recordings per month** and requires a solution that balances transcription accuracy, cost efficiency, and scalability. The evaluation focuses on transcription accuracy using Word Error Rate (WER), cost analysis, and qualitative performance insights.

Accuracy Analysis

The transcription accuracy was evaluated by comparing Whisper's output against a human-verified ground truth transcription. Text normalization (lowercasing, punctuation removal, and whitespace normalization) was applied prior to comparison to ensure a fair evaluation.

Results:

- **Word Error Rate (WER):** 9.78%
- **Accuracy:** 90.22%

Error breakdown:

- Substitutions: 4
- Insertions: 0
- Deletions: 5
- Correct words: 83

Word counts:

- Reference words: 92
- Hypothesis words: 87

Interpretation:

A WER below 10% indicates strong transcription performance for conversational meeting-style audio. Most errors were substitutions and deletions, with no insertions observed, suggesting Whisper tends to omit or slightly alter words rather than hallucinate content. This level of accuracy is suitable for internal documentation, searchable archives, summaries, and meeting notes.

Cost Analysis

Whisper pricing is based on audio duration and is currently **\$0.006 per minute** of processed audio.

Estimated costs for high-volume usage:

- **1,000 hours per month**
 - 60,000 minutes × \$0.006 = **\$360/month**
 - **\$4,320/year**

Even at higher volumes, costs scale linearly and remain predictable, making Whisper highly cost-effective compared to manual transcription or hybrid services.

A more detailed analysis of the costs structure can be found in the last section of this report.

Performance Insights

- Whisper performs well on clear, single-speaker or lightly overlapping meeting audio.
- Errors mainly occur around word substitutions and omissions, which typically do not prevent understanding of the overall content.
- No insertions were observed, indicating low risk of adding incorrect words or fabricated content.
- Accuracy is sufficient for most business use cases without requiring full human review.

Recommendations

Based on the evaluation:

- **Deploy Whisper for large-scale meeting transcription**, especially for internal use cases such as documentation, search, and summaries.
- For **critical or compliance-sensitive meetings**, introduce a lightweight human review process or selectively re-check low-confidence segments.
- Given the low cost and acceptable accuracy, Whisper represents a strong balance between performance and budget for customers processing more than 1,000 hours of recordings per month.

Final recommendation:

Whisper is suitable for production deployment at scale, with optional quality controls depending on the criticality of the content.

Detailed Cost Analysis for High-Volume Usage (1,000+ hours)

We assume the customer has more than 1,000 hours of video content requiring transcription. Based on this assumption, the pricing calculations are outlined below.

Pricing basis

- **Whisper transcription price: \$0.006 per minute**
- Billing is based on **audio duration**, independent of language or content complexity.

Cost calculation

1,000 hours of video per month

- 1 hour = 60 minutes

- 1,000 hours = **60,000 minutes**

Monthly cost

60,000 minutes × \$0.006 = \$360

Annual cost

\$360 × 12 = \$4,320

Cost summary table

Usage volume	Minutes	Monthly cost	Annual cost
100 hours	6,000	\$36	\$432
500 hours	30,000	\$180	\$2,160
1,000 hours	60,000	\$360	\$4,320
2,000 hours	120,000	\$720	\$8,640

Interpretation

- At **enterprise scale**, transcription costs remain **predictable and linear**
- Even at **1,000+ hours/month**, total cost is **well below \$500/month**
- Compared to human transcription or hybrid services, Whisper is **orders of magnitude cheaper**

Cost vs accuracy tradeoff

Based on your WER evaluation:

- If WER is **≤ 10%**, Whisper is suitable for:
 - meeting notes
 - searchable archives
 - summaries

- internal documentation
- For **legal or compliance-critical transcripts**, Whisper output may require:
 - human review
 - selective re-transcription of low-confidence segments

Cost optimization strategies (high-volume)

You can explicitly recommend:

1. **Batch processing** of recordings (no real-time premium)
2. **Optional silence trimming** before transcription (carefully applied)
3. **Tiered quality strategy:**
 - a. Standard meetings → Whisper
 - b. High-risk meetings → Whisper + human review

Recommendation (cost perspective)

From a cost standpoint, Whisper is highly suitable for customers processing more than 1,000 hours of meeting recordings per month. The predictable pricing model and low per-minute cost make it economically viable even at large scale, especially when combined with selective quality controls.